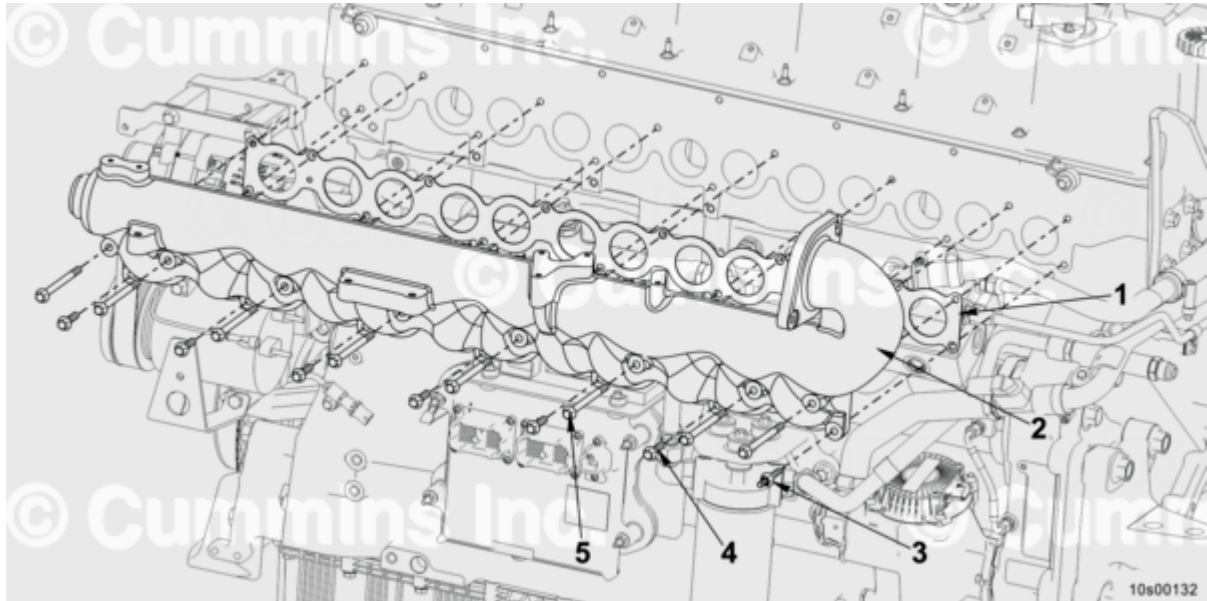


Trainings ▾

Exploded View



Air Intake Manifold Exploded View

1. Gasket - quantity 1
2. Air intake manifold - quantity 1
3. Screw, stud flange cap - quantity 1
4. Screw, hex flange head cap - quantity 7
5. Screw, hex flange head cap - quantity 8

Select Service Tools

Recommended Cummins® Service Tools

- Guide pin, Part Number 3164977

Additional Service Items

- Non-abrasive brush.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

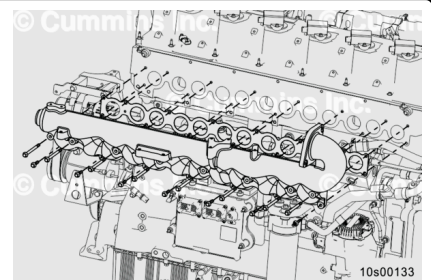
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the charge air piping connection to the air intake connection. See equipment manufacturer service information.
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html>)
- Remove the Venturi tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html>)

Remove

Remove the mounting capscrews securing the air intake manifold to the cylinder head.

Remove and discard the gasket.

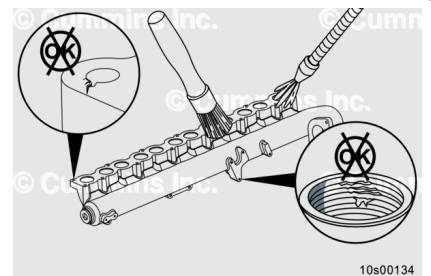
Cover the cylinder head intake ports to prevent debris from entering the engine while the air intake manifold is removed.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the air intake manifold gasket surface.

Clean the air intake manifold. Saturate the inside of the air intake manifold with solvent.

Mineral based solvents can be used if the expected results with the safety solvent are **not** obtained.

Clean the debris with a non-abrasive brush.

Dry the air intake manifold with compressed air.

Inspect the air intake manifold for cracks or other damage.

Replace the air intake manifold if damage is found.

Install

If previously covered, remove the cover from the cylinder head prior to installing the air intake manifold.

Align the air intake manifold with the cylinder head.

Install two guide pins, Part Number 3164977, in the air intake manifold capscrew mounting holes in the cylinder head in the positions shown before installing a new gasket.

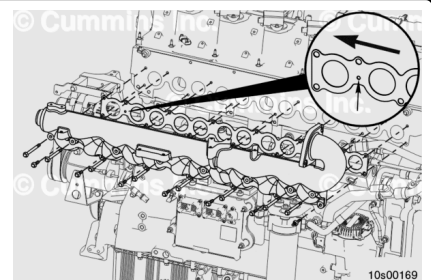
Install a new gasket over the guide pins.

The gasket **must** be oriented with the alignment hole facing the front of the engine.

Install the air intake manifold over the guide pins.

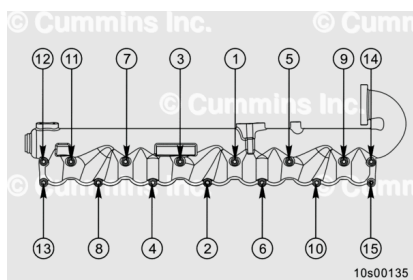
Install the remaining capscrews. Hand tighten the capscrews.

Remove the two guide pins and finish installing the remaining two capscrews hand-tight.



Tighten all the capscrews in the sequence illustrated.

Torque Value: 28 n•m [248 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html)
- Install the charge air piping. See equipment manufacturer service information.
- Install the Venturi tube. Refer to Procedure 011-070 in Section 11.
(/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.